**Reg. No.** | 9 | 5 | 1 | 7 | | | | | | | | |

**Question Paper Code : 21553**

B.E. / B.TECH. DEGREE EXAMINATIONS, NOVEMBER / DECEMBER 2024

Third Semester

Common to Computer Science and Engineering & Information Technology

**23MA303 – MATHEMATICS – III: DISCRETE MATHEMATICS**

(Regulations: Mepco – R2023)

Duration: 3 Hours Max. :100 Marks

Answer ALL Questions

BTL, CO

**PART A – (10 × 2 = 20 Marks)**

R, CO1 1. Construct a truth table for $(\neg p \vee \neg q) \leftrightarrow (p \leftrightarrow q)$.

A, CO1 2. Which of the following statement is logically equivalent to $p \rightarrow q$? Justify your answer.

A) $q \rightarrow p$ B) $q \vee \neg p$ C) $p \wedge \neg q$ D) $p \vee \neg q$

R, CO2 3. Define the chromatic number of a graph. What is the chromatic number of a complete bipartite graph $K_{m,n}$ ?

A, CO2 4. Everybody in a room shakes hands with everybody else. The total number of hand shakes is 66. The total number of persons in the room is ____. Justify your answer.

A) 11 B) 12 C) 33 D) None of these

R, CO3 5. Define a monoid. How does it differ from a semigroup? Provide an example of a monoid.

U, CO3 6. A field is defined as a set that is a _____. Justify your answer.

A) Group under addition and a semigroup under multiplication

B) Monoid under addition and semigroup under multiplication

C) Semigroup under addition and monoid under multiplication

D) None of these

R,CO4 7. When is a lattice said to be (A) bounded (B) distributive?

A,CO4 8. Which of the following is a property of a partially ordered set (poset)? Justify your answer.

A) Reflexivity B) Symmetry C) Transitivity D) Both A and C

A,CO5 9. If $P(A) = 0.9$, $P(B) = 0.6$ and $P(A \cap B) = 0.5$, determine $P(\bar{A} \cap \bar{B})$.

U,CO5 10. For the function $f(x) = a + bx$ for $0 < x < 1$, to be a valid probability density function, which one of the following statements is correct? Justify your answer.

A) a = 1, b = 4 B) a = 0.5, b = 1 C) a = 0, b = 1 D) a = 1, b = -1

**PART B – (4 × 15 = 60 Marks)**

A, CO2 11. a) i. State and prove Euler's formula on planar graphs. (8 Marks)

A, CO2 11. a) ii. Solve the following recurrence relations:

A) $a_{n+2} - 5a_{n+1} + 6a_n = 2$

B) $a_{n+2} - 4a_{n+1} + 4a_n = 2^n$ (7 Marks)

**OR**

A, CO2 11. b) i. In how many ways 5 boys and 4 girls can be seated at a round table if

A) There is no restriction

B) all the four girls sit together

C) all the four girls do not sit together

D) no two girls sit together (8 Marks)

A, CO2 11. b) ii. Find the chromatic polynomials of the following graphs

A) $P_6$ (Path on six vertices)

B) $K_7$ (Complete graph on seven vertices)

C) $C_4$ (Cycle graph on four vertices) (7 Marks)

U, CO3 12. a) i. Show that the set of matrices $\left\{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}\right\}$ form a multiplicative abelian group. (8 Marks)

U, CO3 12. a) ii. Prove the following:

A) Every cyclic group is an abeilan group

B) If $a$ is a generator of a cyclic group $G$, then $a^{-1}$ is also a generator of $G$. (7 Marks)

**OR**

U, CO3 12. b) i. If $f: G \rightarrow G'$ is a group homomorphism from $\{G,*\}$ to $\{G', \Delta\}$, then prove that

A) $f(e) = e'$, where $e$ and $e'$are identity elements of $G$ and $G'$ respectively,

B) for any a$\in G, f(a^{-1}) = [f(a)]^{-1}$ (8 Marks)

U, CO3 12. b) ii. State and prove Lagrange's theorem with the necessary results required to prove it. (7 Marks)

U, CO4 13. a) i. Prove that every chain is a distributive lattice. (8 Marks)

U, CO4 13. a) ii. If $(L, \leq)$ is a lattice, then for any $a, b, c \in L$, prove the following property: If $b \leq c$, then A) $a \vee b \leq a \vee c$ B) $a \wedge b \leq a \wedge c$. (7 Marks)

**OR**

U, CO4 13. b) i. State and prove Modular inequality in a Lattice. (8 Marks)

U, CO4 13. b) ii. In a Boolean algebra, show that

A) $(x + y)(x' + z) = xz + x'y + yz = xz + x'y$

B) $(xy'z' + xy'z + xyz + xyz')(x + y) = x$ (7 Marks)

A,CO5 14. a) i. A discrete random variable $X$ has the probability function given below:

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $P(x)$ | 0 | $k$ | $2k$ | $2k$ | $3k$ | $k^2$ | $2k^2$ | $7k^2+k$ |

Find A) the value of $k$ B) $P(X < 6), P(0 < X < 4)$ and $P(X > 5)$

C) the distribution function of $X$. (8 Marks)

A,CO5 14. a) ii. A random variable $x$ has the p.d.f given by $f(x) = \begin{cases} 2e^{-2x}, x \geq 0 \\ 0, \quad x < 0 \end{cases}$. Find

A) the moment generating function B) the first four moments about the origin. (7 Marks)

**OR**

A,CO5 14. b) i. There are 3 machines producing 10,000, 20,000 and 30,000 syringes per hour respectively. These machines are known to produce 5%, 4% and 2% defective syringes respectively. One syringe is taken at random from an hour's production of the three machines, What is the probability that it is defective? If the drawn syringe is defective, what is the probability that this was produced by the second machine? (8 Marks)

A,CO5 14. b) ii. The amount of bread (in hundreds of pounds) X that a certain bakery is able to sell in a day is found to be a numerical valued random phenomenon, with a probability density function specified by the p.d.f f(x) given by

$$f(x) = \begin{cases} kx, & 0 \leq x \leq 5 \\ k(10 - x), & 5 \leq x \leq 10 \\ 0, & otherwise. \end{cases}$$

Determine the value of $k$. What is the probability that the number of pounds of bread that will be sold tomorrow is A) More than 500 pounds B) less than 600 pounds C) between 250 and 750 pounds (7 Marks)

## PART C – (1 × 20 = 20 Marks)

L, CO1 15. a) i. Show that the premises "It is not sunny this afternoon and it is colder than yesterday", "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will go home by sunset" lead to the conclusion "We will go home by sunset." (10 Marks)

A, CO1 15. a) ii. Prove the following using rules of inference theory

A) Use the indirect method to show that $r \longrightarrow \neg q,\ r \vee s,\ s \rightarrow \neg q,$ $p \longrightarrow q \implies \neg p.$

B) The premises $a \rightarrow (b \rightarrow c), d \rightarrow (b \wedge \neg c)$ and $(a \wedge d)$ are inconsistent. (10 Marks)

**OR**

A, CO1 15. b) i. Prove the following implications by using truth table

$\big(p \rightarrow (q \rightarrow s)\big) \wedge (\neg r \vee p) \wedge q \implies r \rightarrow s$ (10 Marks)

L, CO1 15. b) ii. Prove the following implications

A) $\forall x(P(x) \rightarrow Q(x)),\ \forall x\big(R(x) \rightarrow \neg Q(x)\big) \Longrightarrow \forall x\ \big(R(x) \rightarrow \neg P(x)\big).$

B) A student in this class, knows how to write programs in JAVA. Everyone who knows how to write programs in JAVA can get a high-paying job. Therefore, someone in this class can get a high-paying job. (10 Marks)